

JD for Computer Science & Technology Engineer Trainee

Computer Science & Technology Engineer Trainee is required to bridge the gap between traditional mining operations and modern digital technology. The main focus of CS&T Trainee Engineer will be on implementing software solutions, analysing mining data, automating processes, and supporting technical tools for mine planning and optimization. He/She will work under an AI Expert.

Job Summary

We are looking for Graduate Computer Science & Technology Engineer to join our mining consulting team as a Trainee Engineer. He/She will be responsible for developing, implementing, and maintaining specialised software solutions, IoT systems, and data analytics tools to enhance the efficiency, safety, and productivity of our clients' mining operations. The incumbent will work closely with mining engineers, geologists, and clients to translate technical requirements into scalable, reliable IT solutions.

Key Responsibilities

- **Software Development & Integration:** Design, develop, and deploy custom software applications for mining automation, mine planning, and simulation.
- **Data Analysis & Visualisation:** Collect, process, and analyse vast amounts of data from mining equipment (IoT), geological studies, and production reports to identify optimisation opportunities.
- **Mining Software Support:** Provide technical support and customisation for specialised, industry-standard mining software (e.g., Deswik, Surpac, Vulcan, Maptek, Datamine, MS Project).
- **Automation & AI:** Implement machine learning models and artificial intelligence to improve predictive maintenance, ore quality control, and safety systems.
- **System Optimisation:** Analyse existing operational workflows, troubleshoot, and optimise systems to enhance performance.
- **Technical Documentation & Training:** Prepare detailed user manuals, technical reports, and conduct training for client personnel.
- **Collaboration:** Act as the technical lead in cross-functional teams, bridging the gap between IT infrastructure and on-site mining engineers.

Required Skills & Qualifications

- **Programming Languages:** Strong proficiency in Python, C++, Java, or C#.
- **Database Management:** Expertise in SQL Server, Oracle, or modern cloud data warehouses (e.g., Snowflake).
- **Data Science/ML:** Familiarity with AI/ML frameworks (e.g., TensorFlow, PyTorch) and data visualisation tools (e.g., Power BI, Tableau).
- **Tools:** Experience with CAD software or geospatial data tools is highly desirable.
- **Cloud/DevOps:** Exposure to Azure or AWS cloud services and containerisation tools (Docker, Kubernetes) will be an added advantage.
- **Soft Skills:** Excellent analytical problem-solving skills, strong communication skills, and the ability to work in a collaborative, client-facing environment.

Why to work for a mining consulting firm?

- Work on cutting-edge, industry-transforming technology.
- Direct impact on improving safety and sustainability in mining.
- Opportunities to work on large-scale mining projects.
- Dynamic, hybrid, and inclusive work environment.